Robotics 4/M – 2025

Section A: Sensors, Actuators & Image Processing

Tutorial 1: Sensors Answers

Q1.
a. Define what a Sensor is and provide three examples with applications.

A sensor is a device which detects or measures a physical property and records, indicates or otherwise responds to it. *[Lecture 1]*

Multiple Examples Exist:
Temperature Sensor for detecting the temperature of an environment.
Ultrasonic Sensor for range measurement and obstacle avoidance.
Wheel encoder for positional or velocity information about a rotating joint.
A potentiometer for measuring the angle of a joint.

[As long as a Sensor is named and the application is relevant]

b. The Silicon Sensing Orion Accelerometer has the following specifications:

Selectable Range: ±2.5g or ±10g
Resolution: 0.079mg (±2.5g), 0.313mg (±10g)
Non-linearity: <±3mg (±2.5g), <±5mg (±10g)

Explain the significance of each number provided.

The accelerometer can detect accelerations between ±2.5g or ±10g. *[Lecture 2]*
The minimum detectable change for the ±2.5g range is 0.079mg and for ±10g is 0.313mg.
The non-linearity value indicates the maximum difference between the ideal linear response and the actual value.

c. Given the following specification:

Range: 0-220°C
emf_0: 0mV
emf_{100}: 5.287mV
emf_{220}: 11.436mV

Calculate the non-linearity error at 100°C as a percentage of the full range output, assuming a linear relationship between the emf and temperature over the full range. Work to 3 decimal places.

[Lecture 2] Given: 0-220°C as range.

0°C = 0mv
100°C = 5.287mv
220°C = 11.436mv

Full Range Sensitivity = $\frac{11.436-0}{220-0} = 0.052(3dp)$

Ideal output for 100°C = $100 \times 0.052 = 5.2mV$

Error in output = 5.2-5.287 = -0.087mV

Percentage Error over full scale = $\frac{-0.087}{11.436} \times 100 = -0.761\%$

or $(\frac{-0.087}{11.436}) = -0.008, -0.008 \times 100 = -0.8\%$*[Sticking strictly to 3dp!]*

Q2.

Explain what is sensor range, resolution, dead band and sensitivity.

Range defines the limits between which the input can vary. Temperature Sensor Range -45°C to +150°C
Resolution is the smallest change in the input value that will result in a change in the output value
Range of the input variable for which the output remains zero

Sensitivity: Relationship between the output and input: But can also be used to describe effect on input from environmental changes

Q3.
A potentiometer is used as an angular position sensor for a rotating joint in a robotic arm. The maximum resistance, R, of the potentiometer is 500Ω. When the joint is at 0° the resistance, R_p, is 100Ω, at 22.5° R_p is 175Ω, at 45° R_p is 250Ω, at 67.5° R_p is 325Ω and at 90° R_p is 400Ω. Work to 3 decimal places.

a. What is the resistance when the joint is at an angle of 60°?

[Lecture 2]

Full Range Sensitivity = $\frac{400-100}{90-0} = 3.333\ \Omega/°$

Value of Rp @ 60° = $(60 \times 3.333) + 100 = 299.980\Omega$

b. With a Vs of 10V and no load, what are the output voltages for 0°, 45° and 90°?

Noting that $V_o = x \times V_s$ where $x = \frac{R_p}{R}$

$For 0° \Rightarrow V_o = \frac{100}{500} \times 10 = 2V$

$For 45° \Rightarrow V_o = \frac{250}{500} \times 10 = 5V$

$For 90° \Rightarrow V_o = \frac{400}{500} \times 10 = 8V$

c. With a load of 5KΩ, what are the output voltages for 0°, 45° and 90°?

Noting that $\boldsymbol{V_O = V_S.(\frac{x}{\frac{R}{R_L}.x.(1-x)+1})}$

$For 0° \Rightarrow V_S = 10V, x = \frac{100}{500} = 0.2, R_P = 100\Omega, R_L = 5k\Omega, \frac{R}{R_L} = 0.1$

$V_O = 10.(\frac{0.2}{0.1*0.2.(1-0.2)+1}) = 10.(\frac{0.2}{0.02(0.8)+1}) = 10.(\frac{0.2}{1.016}) = 1.969V$

$For 45° \Rightarrow V_S = 10V, x = \frac{250}{500} = 0.5, R_P = 250\Omega, R_L = 5k\Omega, \frac{R}{R_L} = 0.1$

$V_O = 10.(\frac{0.5}{0.1*0.5.(1-0.5)+1}) = 10.(\frac{0.5}{0.05(0.5)+1}) = 10.(\frac{0.5}{1.025}) = 4.878V$

$For 90° \Rightarrow V_S = 10V, x = \frac{400}{500} = 0.8, R_P = 400\Omega, R_L = 5k\Omega, \frac{R}{R_L} = 0.1$

$V_O = 10.(\frac{0.8}{0.1*0.8.(1-0.8)+1}) = 10.(\frac{0.8}{0.08(0.2)+1}) = 10.(\frac{0.8}{1.016}) = 7.874V$

d. What is the full-scale error, as a percentage, in the output voltages for 0°, 45° and 90° with a load of 5KΩ?

Noting that $\boldsymbol{error = V_S.\frac{R_P}{R_L}.(x^2 - x^3)}$

$$For 0° \Rightarrow error = 10.\frac{500}{5000}.(0.2^2 - 0.2^3) = 10 \times 0.1 \times (0.04 - 0.008) = 0.032$$
$$\Rightarrow (\frac{0.032}{10}) \times 100 = 0.32\%$$

$For 45° \Rightarrow error = 10.\frac{500}{5000}.(0.5^2 - 0.5^3) = 10 \times 0.1 \times (0.25 - 0.0625) = 0.1875 \Rightarrow (\frac{0.1875}{10}) \times 100 = 1.875\%$

$For 90° \Rightarrow error = 10.\frac{500}{5000}.(0.8^2 - 0.8^3) = 10 \times 0.1 \times (0.64 - 0.512) = 0.128 \Rightarrow (\frac{0.128}{10}) \times 100 = 1.280\%$

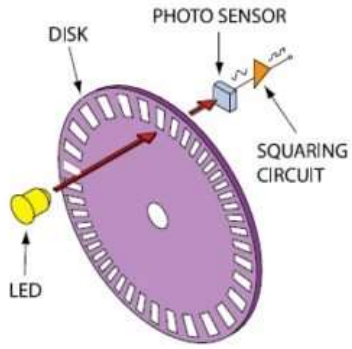
Q4.

a. What is the difference between an *Absolute Encoder* and an *Incremental Encoder*?

An Absolute Encoder returns an actual angular position whereas an incremental encoder detects changes in position. The Absolute encoder achieves this by having a unique code for every position of the shaft.

b. Explain the basic working principle of an optical encoder. A figure should be used.

The basic principal of operation is that a disk with either cut-outs or etched lines rotates as the shaft it is on rotates. A light source is shone onto the disk with a light detector on the opposite side. As the disk rotates the light beam is broken by the solid areas of the disk. This produces a pulse on the detector that is used to indicate a change in position occurred. As the disk rotates a pulse stream is formed. The pules can be counted to indicate position change.



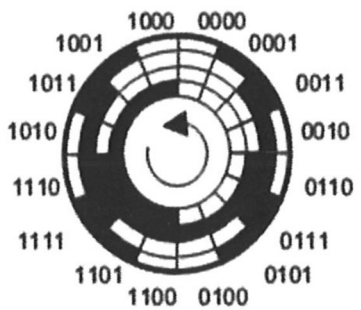
c. If an encoder has 8 tracks, what is the achievable resolution to 3 decimal places?

$$8\ tracks, so\ 8bits, giving\ 2^8 = 256, resolution = \frac{360°}{256} = 1.406°$$

d. On a single-track encoder, how many lines (or cutouts) are required to achieve a resolution of 0.1°?

$$1track, 0.1° required so \frac{360}{0.1} = 3600$$

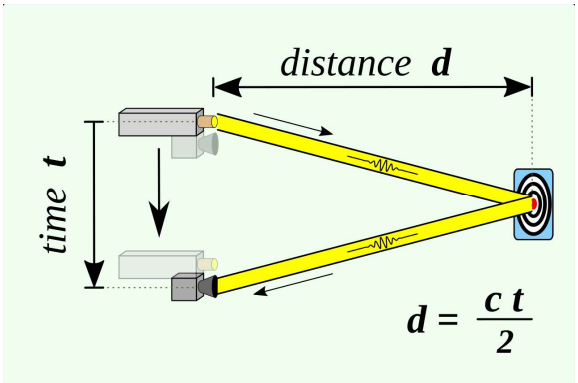
e. Draw the disc layout of a 4-track gray code encoder



Q5.
a. Describe the Time of Flight method for range finders. A figure should be used.

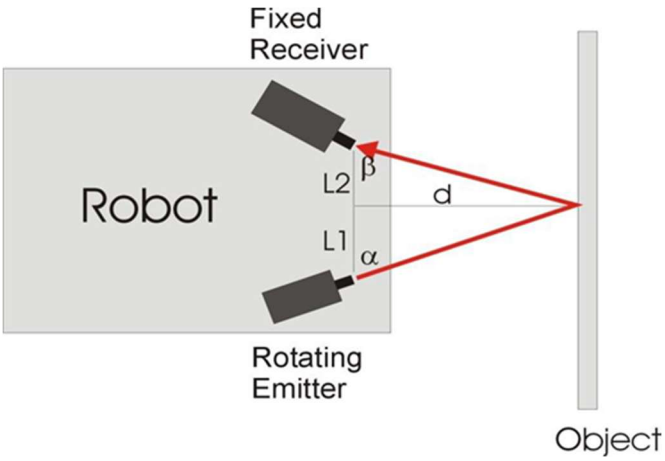
Time of flight range finding consists of sending a signal from a transmitter that reflects off of an

object is detected by the receiver The distance between the sensor and object is half the distance travelled by the signal and is calculated from the time of travel and speed of the signal.



b. Describe the Optical Triangulation method for range finding. A figure should be used.

Triangulation involves illuminating an object by a single ray of light that forms a spot on the object. The spot is detected by the receiver and the range, or depth, is calculated from the triangle formed between the receiver, light source and spot on the object.



c. An Ultrasonic range finder is being used and an echo is detected after 500µs. How far away is the object? (note the speed of sound is 343ms^{-1})

$$s = v.t \Rightarrow Distance = \frac{(343 \times 0.0005)}{2} = 0.08575m$$

d. With respect to Optical Triangulation, given L at 50cm, α = 80° and an object distance of 0.425m. What is the angle β?

$$d = \frac{L.\tan\alpha.\tan\beta}{\tan\alpha + \tan\beta}$$

$$0.425 = \frac{0.5 \times \tan(80) \times \tan\beta}{\tan(80) + \tan\beta}$$

$$0.425 = \frac{0.5 \times 5.671 \times \tan\beta}{5.671 + \tan(\beta)}$$

$$0.425 = \frac{2.836 \times \tan\beta}{5.671 + \tan(\beta)}$$

$$(0.425 \times 5.671) + 0.425 \times tan(\beta) = 2.836 \times tan(\beta)$$

$$2.410 = 2.836 \times tan(\beta) - 0.425 \times tan(\beta)$$

$$2.410 = 2.411 \times tan(\beta)$$

$$tan(\beta) = \frac{2.410}{2.411}$$

$$\beta = \tan^{(-1)}(\frac{2.410}{2.411})$$

$$\beta = 44.988°$$

Q6.

a. **What will be the change of resistance of an electrical resistance strain gauge with a gauge factor of 2.25 and resistance of 50 Ω if it is subjected to a strain of 0.002? Use equation 1.**

$$\boldsymbol{\frac{\Delta R}{R} = K.\frac{\Delta L}{L} = K.\varepsilon} \qquad \textbf{(1)}$$

$$\frac{\Delta R}{R} = K.\frac{\Delta L}{L} = K.\varepsilon$$

$$K = 2.25, R = 50\Omega, \varepsilon = 0.002$$

$$\frac{\Delta R}{R} = K.\varepsilon$$

$$\Delta R = K.\varepsilon.R$$

$$\Delta R = 2.25 \times 50 \times 0.002$$

$$\Delta R = 0.225\Omega$$

b. **A metal rod has an axial length of 200mm. A strain gauge with a gauge factor of 2.2 and a nominal resistance of 250Ω is attached to the rod to measure axial strain. An axial tensile load is applied to the rod. The single strain gauge is connected into a Wheatstone bridge circuit with a supply voltage of 10V. The output voltage from the bridge changes by 0.02V.**

What is the new length of the rod? Use equation 2.

$$\boldsymbol{V_O = V_S \cdot \left(\frac{R_2}{R_1+R_2} - \frac{R_x+\Delta R}{R_x+R_3+\Delta R}\right)} \qquad \textbf{(2)}$$

$$V_S = 10V, R = 250\Omega = R_1 = R_2 = R_3 = R_x, K = 2.2, V_O = 0.02V$$

$$0.02 = 10 \times \left(\frac{250}{250+250} - \frac{250+\Delta R}{250+250+\Delta R}\right)$$

$$\frac{0.02}{10} = 0.5 - \frac{250+\Delta R}{500+\Delta R}$$

$$0.002 - 0.5 = \frac{-250+\Delta R}{500+\Delta R}$$

$$\frac{250+\Delta R}{500+\Delta R} = 0.498$$

$$250 + \Delta R = 0.498.(500 + \Delta R)$$

$$250 + \Delta R = 249 + (0.498.\Delta R)$$

$$\Delta R - 0.498.\Delta R = 249 - 250$$

$$0.502.\Delta R = -1$$

$$\Delta R = \frac{-1}{0.502}$$

$$\Delta R = -1.992\Omega$$

$$\frac{\Delta R}{R} = K.\frac{\Delta L}{L} = K.\varepsilon$$

$$K = 2.2, R = 250\Omega, \Delta R = -1.992\Omega, L = 200$$

$$\frac{\Delta R}{R} = K.\frac{\Delta L}{L}$$

$$\frac{-1.992}{250} = 2.2.\frac{\Delta L}{200}$$

$$-0.008 = 2.2.\frac{\Delta L}{200}$$

$$-0.004 = \frac{\Delta L}{200}$$

$$\Delta L = -0.004 * 200$$

$$\Delta L = -0.8$$

$$newL = 200 - 0.8$$

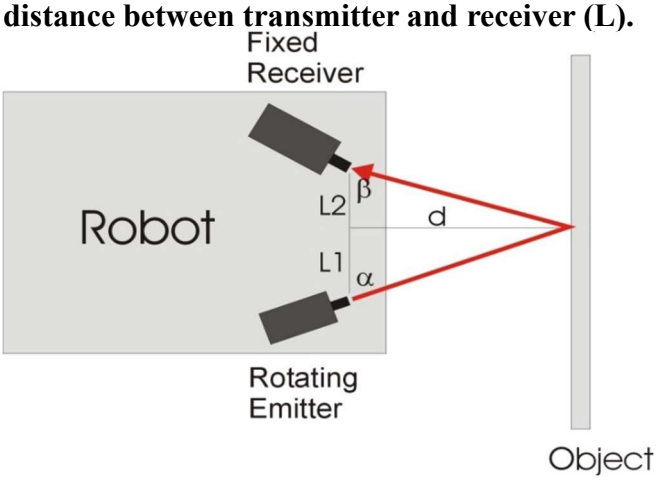
$$L = 199.2mm$$

Past Paper questions

2007 Q1

(e) A robotic rover vehicle uses an optical triangulation system for range finding: [7]

(i) Sketch this system indicating the Transmitter angle (α), Receiver angle (β) and the distance between transmitter and receiver (L).



(ii) Derive from first principles an expression for the object distance in terms of α, β and L.

The following relations can be used to obtain object distance, d:

$$\tan\alpha = \frac{d}{L_1} \tan\beta = \frac{d}{L_2}$$

$$L_1 = \frac{d}{\tan\alpha} L_2 = \frac{d}{\tan\beta}$$

The total distance between the emitter and receiver, L, is known and we can therefore find the distance from:

$$L = L_1 + L_2 = \frac{d}{\tan\alpha} + \frac{d}{\tan\beta} = \frac{d.tan\beta + d.tan\alpha}{\tan\alpha.\tan\beta}$$

$$d = \frac{L.tan\alpha.\tan\beta}{\tan\alpha + \tan\beta}$$

(iii) Given that a 1000 line incremental encoder is used to measure transmitter angle, determine the resolution in distance at a range of 2m, for β = 85° and L = 50cm.

Start with working out α

$$2 = \frac{0.5 \times tan\alpha . tan(85°)}{tan\alpha + \tan(85°)}$$

$$2 \times tan\alpha + 2 \times tan(85°) = 0.5 \times tan\alpha . tan(85°)$$

$$2 \times tan(85°) = 0.5 \times tan\alpha . tan(85°) - 2 \times tan\alpha$$

$$22.86 = 0.5 \times tan\alpha . tan(85°) - 2 \times tan\alpha$$

$$22.86 = tan\alpha(0.5 \times tan(85°) - 2)$$

$$22.86 = 3.715 tan\alpha$$

$$\alpha = \tan^{-1} \frac{22.86}{3.715}$$

$$\alpha = \tan^{-1} 6.153$$

$$\alpha = 80.769°$$

$$encoderresolution = 360/1000 = 0.36\,°/line\ nextangleposition = 80.769° + 0.36° = 81.129°$$

$$Calculate\ the\ distance\ for\ the\ new\ angle\ldots$$

$$d = \frac{L . tan\alpha . \tan\beta}{\tan\alpha + \tan\beta}$$

$$d = \frac{0.5 . \tan 81.129 . tan 85}{\tan 81.129 + tan 85}$$

$$d = \frac{36.617}{17.837}$$

$$d = 2.052m$$

$$Answer = 2.052 - 2 = 0.052m$$